

Engineering Innovation Challenge 2026

SentinelDepth: Privacy-Preserving Edge AI Surveillance for Sustainable Smart Monitoring

Project Code: J-36

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Project Abstract

Conventional surveillance streams and stores identifiable RGB video, creating privacy risk, a wide attack surface, and rising cloud-energy demand. SentinelDepth reframes the device: not another camera watching you, but a companion that keeps you safe. It runs anomaly detection, action recognition, and an on-device assistant entirely on low-power edge hardware, using depth maps and skeletal keypoints instead of continuous RGB video.

The prototype is a single \$200 unit: a Raspberry Pi 5 with a Hailo AI accelerator, an Intel RealSense depth sensor, a 7-inch touch screen, and a microphone and speaker. A quantized convolutional autoencoder flags anomalous depth frames using a per-scene threshold, while MoveNet and an EfficientGCN graph network classify 25 actions into four threat levels, from routine activity to interpersonal aggression and medical distress. No identifiable footage is kept in routine operation; a short clip is stored locally only for high-risk events.

A detected fall triggers help, not a recording: the resident taps “I’M OK” or “GET HELP,” and if there is no response within 30 seconds a caregiver’s phone is alerted. An on-device language model lets the system talk back, offering company as well as safety for people living alone.

On a 580-frame set the autoencoder reached 98.8% accuracy with zero false positives; EfficientGCN reached 80.2% Top-1 (96.0% Top-5) across 25 actions. The pipeline runs today on the \$200 device with a 4.7 MB model, about 2-second response, and zero bytes sent to the cloud—deployment-ready for nursing homes, clinics, schools, and smart buildings.

Project Report

1 Background and Purpose

The global video surveillance market is estimated at USD 71.7 billion in 2026 and is projected to reach USD 118.8 billion by 2031 [1]. Reviews of randomized and natural experiments find that cameras reduce property crime [2]. This protection has a cost. Conventional systems record identifiable RGB video around the clock and often store it in the cloud. The risks are documented. The U.S. Federal Trade Commission found that Ring employees viewed customers' home footage without authorisation and that attackers took over user cameras [3]. In Singapore, the Personal Data Protection Commission fined Marina Bay Sands S\$315,000 in October 2025 after data of 665,495 patrons was stolen and offered for sale on the dark web [4]. The Personal Data Protection Act requires organisations to minimise and protect the personal data they collect [5]. Constant monitoring also changes how people behave [6].

Cloud analytics has an energy cost as well. Data centres consumed about 415 TWh of electricity in 2024, and the International Energy Agency projects about 945 TWh by 2030 [7]. A system that never streams video off the device avoids this load and supports more sustainable monitoring.

Current privacy-preserving monitors use radar, PIR or neuromorphic sensors [8]. They avoid optical imagery, but they detect only single events such as falls and produce no reviewable evidence. RGB analytics understand scenes well but expose identity. Depth and skeleton data carry much lower re-identification risk [9]. SentinelDepth sits between the two approaches. It runs depth-based anomaly detection and skeleton-based action recognition on one edge device [10], keeps a short evidence clip only when a high-risk event occurs, and adds an interactive emergency response and an on-device assistant. In eldercare especially, the same unit can reassure and keep company with residents living alone, rather than only watching them. This report shows that the prototype works in real time on low-cost hardware.

2 Research Questions, Hypotheses, Engineering Goals and Expected Outcomes

Research questions. Can depth-first sensing detect anomalies accurately while preserving privacy? Can a lightweight graph network recognise actions on embedded hardware? Can edge-only inference remove routine cloud transmission at real-time speed?

Hypotheses. **H1:** a depth autoencoder calibrated to its scene separates normal from anomalous events with zero false alarms. **H2:** EfficientGCN-B0 fine-tuned on local data classifies 25 actions above 75% Top-1 accuracy. **H3:** the full pipeline runs continuously on a Raspberry Pi 5 with a Hailo accelerator in under 5 MB of models and retains no identifiable

footage in normal operation.

Engineering goals. Integrate depth anomaly detection, skeleton-based action recognition, an interactive fall/emergency-response flow, and an on-device conversational assistant into a single self-contained edge unit costing about \$200 that sends no data off-device in routine operation.

Expected outcomes: a working prototype, held-out evaluations of both models, and measured footprint, response time and cost on the device.

3 Prototype Design and Methodology

Hardware (Fig. 1). The prototype is a single self-contained unit costing about \$200. A Raspberry Pi 5 (8 GB) runs the system, with a Hailo AI accelerator handling neural inference. An Intel RealSense depth sensor provides the depth stream and captures a short RGB evidence clip only when required. A 7-inch touch screen shows on-site status and hosts the resident-facing prompts, and a microphone and speaker support the on-device assistant. A microSD card holds the operating system, models, logs and any retained clips.

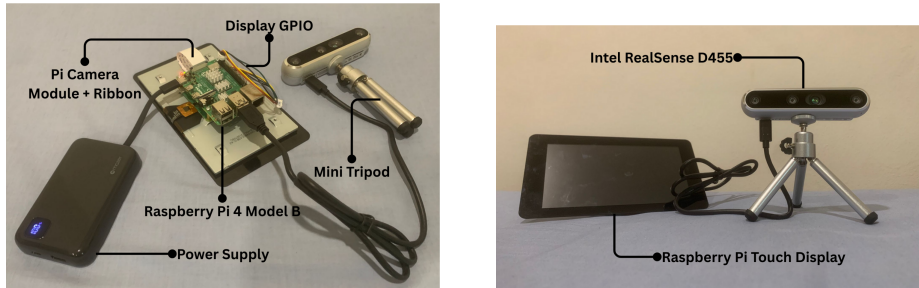


Figure 1: Prototype hardware. Left: top view with the compute board, display and power supply. Right: front view with the Intel RealSense depth sensor.

Depth anomaly detection (Fig. 2). The depth sensor streams 16-bit depth frames at up to 1280×720 . Frames are normalised to $[0, 1]$, downsampled to the network input size, and stacked three at a time so the model sees short-term motion. A quantized convolutional autoencoder reconstructs each stack. The reconstruction error gives a deviation score, following the standard reconstruction-based approach [11, 12]. The autoencoder was trained on TIMo, an Azure Kinect indoor monitoring dataset [13]. Reconstruction detectors are sensitive to scene layout, so the threshold is calibrated per scene: the camera records 10 to 20 seconds of known-normal footage while static, and the threshold is set with a safety margin against the observed normal scores. This calibration is currently a manual setup step. A one-touch routine is planned.

Action recognition (Fig. 2). A transient frame held in volatile memory feeds MoveNet Lightning, which extracts 17 body keypoints in real time [16]. The frame is never written to storage. The classifier does not depend on where the joints come from: EfficientGCN-B0 [15] reads joint coordinates only. V2V-PoseNet [17] and A2J [18] show that the same joint data can be recovered from depth maps alone, so a future version can drop RGB input entirely (§6).

Keypoint sequences map to 25 actions from the NTU RGB+D 120 taxonomy [14] and then to four threat levels (Table 1). A Level 2 or higher event, or an anomaly trigger, retains a 10–15 s clip in local storage for authorised review. Everything else is discarded. No data leaves the device.

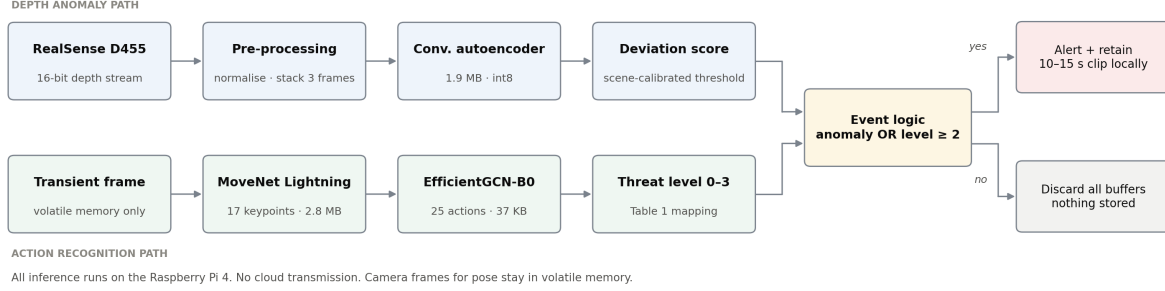


Figure 2: Processing pipeline. All inference runs on the device. Identifiable imagery exists only in volatile memory unless the event logic triggers local retention.

Threat level	System response	Monitored actions (25 classes)
0: Safe (8)	None; buffers discarded	drink water; eat meal/snack; sitting down; standing up; reading; writing; typing on a keyboard; make/answer phone call
1: Suspicious (5)	Attention flag in event log	reach into pocket; pointing with finger; move heavy objects; take something out of a bag; arm circles
2: High, aggression (8)	Alert; retain 10–15 s clip for authorised review	punching/slapping; kicking other person; pushing other person; hit other person with something; wield knife; knock over other person; shoot with a gun; put something into a bag
3: Critical, distress (4)	On-screen “I’M OK / GET HELP” prompt; caregiver alerted if no reply in 30 s; retain 10–15 s clip	falling; staggering; touch chest (heart pain); nausea/vomiting

Table 1: The 25 monitored action classes and their threat levels (taxonomy from NTU RGB+D 120 [14]).

Emergency response and on-device assistant. A fall or other Level 3 event triggers help rather than a recording. The touch screen shows an “I’M OK” / “GET HELP” prompt; if the resident does not respond within 30 seconds, or selects “GET HELP,” a caregiver’s phone is alerted. In parallel, an on-device large language model provides a conversational assistant, so a resident living alone gains company as well as monitoring. Speech is processed locally and no audio leaves the device.

Training and deployment. EfficientGCN-B0 was pretrained on NTU RGB+D 120 [14] and fine-tuned on 641 locally captured skeleton sequences covering the 25 actions (70:20:10 split), with sequence-level augmentation to offset the small dataset. Training ran offline on a laptop. Both models were quantized to int8 with TensorFlow Lite and deployed for on-device, Hailo-accelerated inference.

4 Procedures, Risk and Safety, and Methods of Data Analysis

Action sequences were recorded from teammates and student volunteers with informed consent. Only skeletal keypoints were kept for training. Audio for the assistant is processed on-device and is never transmitted. The prototype runs at 5 V from a USB power bank. The depth sensor’s infrared projector is a Class 1 eye-safe laser. All equipment was bench-mounted. The anomaly threshold was calibrated from known-normal footage as described in §3. Detection is reported as accuracy, precision, recall, F1 and ROC-AUC. Action recognition is reported as Top- k accuracy and per-class F1 on the held-out split.

5 Results and Evaluation

Depth anomaly detection (Fig. 3, left). The evaluation set contains 580 frame samples from the deployment scene: 240 normal and 340 anomalous. The deviation scores of the two groups were fully separated, with a clear gap between their ranges. The calibrated threshold sits on the normal side of that gap with a safety margin. It produced zero false positives and seven false negatives: 98.8% accuracy, 100% precision, 97.9% recall, F1 of 99.0%, AUC $\approx$ 1.00. The margin trades seven borderline anomalies for headroom against false alarms, which are the main failure cost in deployed monitoring. This set is small and scene-specific; broader continuous-operation testing across scenes and calibration modes remains future work.

Action recognition (Fig. 3, right). The fine-tuned EfficientGCN reaches 80.16% Top-1, 92.86% Top-3 and 96.03% Top-5 accuracy on the held-out split, from skeleton data only. Falling, reading, typing and weapon-related actions reach F1 of 1.00. Actions involving two people score lowest: putting an object into a bag at 0.40, and punching or slapping at 0.46. This matches the single-person pose extractor. Per-class support is small, so per-class figures are indicative. The published EfficientGCN-B0 reaches about 90% cross-subject accuracy on NTU RGB+D 60 with far more training data [15]. 80% Top-1 from 641 sequences is consistent with that gap.

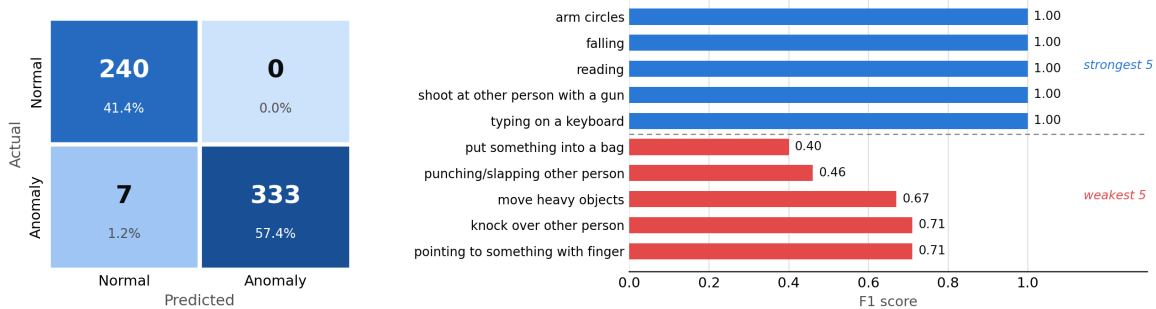


Figure 3: Left: anomaly detector confusion matrix on the 580-sample set. Right: strongest and weakest action classes by F1 on the held-out split.

System performance against industry practice. The model stack totals 4.7 MB: au-

toencoder 1.9 MB, MoveNet 2.8 MB, EfficientGCN 37 KB. On the Raspberry Pi 5 with the Hailo accelerator the system delivers an end-to-end response of about 2 seconds while running entirely on-device. Commercial radar fall detectors avoid cameras, but they detect only falls and presence, produce no reviewable evidence, and offer no interactive response or companion function. A streaming 1080p IP camera uploads 4 to 8 GB per day; SentinelDepth uploads nothing in routine operation, at a total hardware cost of about \$200.

6 Limitations

Depth sensing degrades under strong outdoor infrared light. MoveNet Lightning tracks one person, which limits crowded scenes and explains the weakest action classes. The anomaly detector needs per-scene threshold calibration. Broader normal-scene training data is planned to improve transfer across scenes and to unseen anomaly types. Pose extraction still uses transient camera frames; a depth-domain pose model [17, 18] would remove them. The emergency-response flow and on-device assistant are implemented but not yet quantitatively evaluated for response reliability, false-trigger rate, or conversational accuracy. Both model evaluation sets are small and controlled. Results show feasibility, not field performance.

7 Conclusion and Future Work

SentinelDepth analyses depth and skeletal data entirely on a low-cost (\$200) edge device built around a Raspberry Pi 5 and a Hailo accelerator. Identifiable imagery exists only in volatile memory unless a high-risk event triggers a short local clip. On its evaluation sets the prototype detected anomalies with zero false positives and classified 25 actions at 80% Top-1 accuracy, within 4.7 MB of models and about a 2-second response, sending zero bytes to the cloud. Beyond monitoring, a fall triggers an interactive help prompt with caregiver escalation, and an on-device assistant offers company for residents living alone. The design fits privacy-sensitive settings such as nursing homes, clinics, schools and offices, reframing the device from a camera that watches into a companion that keeps people safe.

Planned work: automatic per-scene calibration, retraining on broader normal footage, a depth-domain pose estimator, multi-person tracking, quantitative evaluation of the emergency-response flow and on-device assistant, and an infrared calibration or LiDAR fallback for bright environments.

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Declaration of Generative AI Use

AI tools were used only for code debugging and language refinement of the report, research plan and abstract. All experiments, measurements, analyses, planning, and reported results were conducted by our team. No AI tools were used to fabricate, manipulate, or alter any data, findings, or conclusions.